

Antigravity IDE & Runtime Setup

The Production Environment Configuration Blueprint

Executive Summary: This document provides actionable production implementations engineered by Pinnacle AI Solutions. Use this blueprint to architect zero-downtime, autonomous AI agent pipelines that streamline operational scale and client acquisition.

1. The Autonomous Agentic Runtime Environment

Google DeepMind's Antigravity represents a paradigm shift from passive code completions to proactive agentic pair-programming. Unlike traditional IDE assistants that simply suggest syntax, Antigravity functions as an autonomous multi-modal agent with native terminal orchestration, browser automation, and MCP protocol extensibility.

To unlock maximum performance on Windows 11 development machines, agents require dedicated PowerShell execution profiles, non-interactive execution guards, and localized scratch environments.

- Unified Execution Shell: Native PowerShell 7 (pwsh) configured with strict non-interactive flags.
- Scratch Sandboxes: Dedicated artifact and scratch directories mapped outside version-controlled working trees.
- Self-Healing State Management: Automatic recovery via structured transcripts and background task supervision.

2. System Configuration & Environment Bootstrapping

Deploy the core configuration profile to `~/.gemini/config/GEMINI.md` or workspace `.agents` rules. Ensure PATH variables prioritize isolated Python virtual environments and high-speed package runtimes.

- Set `PAGER=cat` to eliminate terminal blocking on long command outputs.
- Verify primary workspace as `C:\Users\futur\gemini_workspace`.
- Enforce atomic log rolling to prevent disk saturation during continuous agent runs.

```
# Initialize Isolated Agentic Workspace Profile
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
pwsh -NoProfile -Command "
    Write-Output 'Antigravity Workspace Initialized'
    $env:PAGER = 'cat'
    $env:PYTHONUNBUFFERED = '1'
"
```

3. Operational Rules & Safety Gate Doctrines

Enterprise reliability requires strict boundaries. All agentic modifications must adhere to non-destructive refactoring principles. Never permit autonomous deletion of core assets without automated snapshot backups and explicit user confirmation.

- Pre-Action Verification: Mandatory validation before modifying configuration roots.
- Persistent Lead Retention: Continuous dual-write to both `leads.json` and `leads.csv`.

- Database Fallbacks: Zero-latency synchronization to Neon PostgreSQL and Supabase.